

NYC Jobs Market Intelligence Pipeline

End-to-end data engineering + NLP project: turned 2,148 raw NYC government job postings into a sector-classified, statistically-audited market intelligence report — with zero manual labeling and zero data sent to any third-party AI service.

2,148 RAW POSTINGS PROCESSED	8 SECTORS · ZERO-SHOT NLP	100% LOCAL / OFFLINE · NO 3RD-PARTY API	4 PIPELINE STAGES (SQL-BACKED ETL)	6 STATISTICAL SAFEGUARDS
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■ Data Engineering

Built a 4-stage SQL-backed pipeline — raw CSV ingestion → MySQL warehouse → NLP sector classification → feature engineering → statistical analysis — with every intermediate table persisted and independently auditable.

Engineered 5 derived features — Avg_Salary, Salary_Flexibility_%, Executive flag, Days_To_Apply, and a standardized 3-tier seniority scale from messy civil-service level codes, with defensive divide-by-zero handling.

■ NLP & Semantic Classification

Cleaned ~2,000 noisy job titles — using spaCy POS-tag filtering (NOUN/PROPN extraction) instead of a brittle keyword list, so it generalizes to titles never seen before.

Classified postings into 8 sectors with zero labeled data — via Sentence-Transformer embeddings (all-MiniLM-L6-v2) and weighted cosine similarity (title 0.7 / description 0.3), avoiding the cost of manual ground-truth tagging.

Evaluated 3 competing approaches (TF-IDF, hosted LLM API, local embeddings) — and chose deliberately on speed, semantic accuracy, and data privacy — rejecting the LLM route specifically because it would send every posting to a third-party endpoint.

■ Statistical Judgement

Self-audited every headline number — flagging mean-vs-median skew, recommending ANOVA/Kruskal-Wallis significance tests and Pearson/Spearman correlation before treating any pattern as confirmed.

Deliberately withheld predictive modeling — recognized that strata as small as n=1–12 (e.g. Marketing & Sales, Senior-level Data roles) can't support a trained model without curve-fitting to noise — a judgment call over a flashier but less honest deliverable.

■ Output

8-page business intelligence report — sector pay benchmarks, seniority/executive pay gaps, in-demand titles, and a dedicated data-roles market spotlight.

9-page technical methodology appendix — documenting every pipeline decision for reproducibility and review.

■ Tech Stack

Python	pandas / NumPy
SQLAlchemy + MySQL	spaCy (NLP)
Sentence-Transformers	scikit-learn
Statistical Testing	Data Visualization